BVAR Scenario Tool
User Manual and Methodological Documentation[1]

Mátyás Farkas

International Monetary Fund

June 10, 2026

Source code: https://github.com/matyasfarkas/BVARScenarioTool

[1]The views expressed herein are those of the author and should not be attributed to the International Monetary Fund, its Executive Board, or its management. This tool is provided "as is" without warranty of any kind, express or implied. The author and the IMF disclaim all liability for any errors, omissions, or for any loss or damage arising from its use. Users are solely responsible for any decisions made based on the output of this tool.

BVAR Scenario Tool
User Manual and Methodological Documentation

Mátyás Farkas
International Monetary Fund

June 10, 2026

Abstract

This document describes the BVAR Scenario Tool, a browser-based application for conditional forecasting, scenario analysis, and policy counterfactual evaluation. The tool combines a large Bayesian vector autoregression (BVAR) estimated with hierarchical Minnesota priors (Giannone et al., 2015) with structural impulse responses drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016). Users condition the BVAR forecast on arbitrary variable paths via a Durbin–Koopman simulation smoother (Durbin and Koopman, 2002, 2012), and evaluate monetary and fiscal policy counterfactuals using DSGE-identified impulse response functions. We describe the methodology employed, the implementation of the conditional forecasting algorithms, and the three-layer architecture that separates the unconditional baseline, the user's scenario, and the policy counterfactual. All computation runs client-side in the browser; the tool requires no server infrastructure and can be deployed as a static website.

Keywords: BVARs, Conditional Forecasting, Scenario Analysis, Kalman Filter, Durbin–Koopman Smoother, Policy Counterfactuals, DSGE Models, MMB

JEL classification: C32, C53, E37, E52, E58

---

*The views in this paper are solely the responsibility of the author and should not be interpreted as reflecting the views of the International Monetary Fund or any other person associated with the IMF. Email: `mfarkas@imf.org`.

Contents

1 Introduction

Conditional forecasting with Bayesian vector autoregressions (BVARs) has become a standard tool in central bank forecasting and policy analysis. The seminal contributions of Doan et al. (1984) and Waggoner and Zha (1999) established the methodological foundations, while Bańbura et al. (2015) extended these techniques to large cross-sections using state-space methods. The BVAR Scenario Tool described in this document builds on this literature to provide an interactive, browser-based platform that makes conditional forecasting and policy counterfactual analysis accessible without specialized software.

The tool implements a three-layer analytical framework:

(i) **Baseline forecast:** An unconditional forecast from a large BVAR estimated on quarterly macroeconomic and financial data, following the hierarchical prior framework of Giannone et al. (2015).

(ii) **Conditional scenario:** The user conditions any subset of variables on desired paths; a Durbin–Koopman smoother (Durbin and Koopman, 2002, 2012) propagates these constraints to all remaining variables, respecting the full covariance structure of the VAR.

(iii) **Policy counterfactual:** An alternative monetary or fiscal policy path is overlaid on the scenario using impulse response functions from a DSGE model drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012), and the resulting macroeconomic responses are re-conditioned through the BVAR to ensure cross-variable consistency.

The tool is designed for economists at central banks and international institutions who need to construct and communicate conditional projections and policy scenarios. It runs entirely in the browser, requiring no server-side computation, and can be deployed as a static website on any hosting platform.

This document is organized as follows. Section 2 describes the BVAR estimation methodology. Section 3 derives the companion-form state-space representation. Section 4 presents the Kalman filter and Durbin–Koopman smoother used for conditional forecasting. Section 5 details the policy counterfactual framework. Section 9 documents the variable coverage. Section 10 provides a guide to the user interface. Section 11 discusses technical implementation details. Section 12 concludes with a discussion of limitations and directions for future development.

2 BVAR Estimation

2.1 Reduced-Form VAR

Consider an n-variable VAR(p) in reduced form:

$$y_t = c + \sum_{\ell=1}^{p} B_\ell \, y_{t-\ell} + u_t, \qquad u_t \sim \mathcal{N}(0, \Sigma), \tag{1}$$

where $y_t \in \mathbb{R}^n$ is the vector of observables, $c \in \mathbb{R}^n$ is a vector of intercepts, $B_1, \ldots, B_p \in \mathbb{R}^{n \times n}$ are the autoregressive coefficient matrices, and $\Sigma \in \mathbb{R}^{n \times n}$ is the positive-definite innovation covariance matrix. In the US specification, $n = 36$ and $p = 3$, yielding a coefficient matrix $\hat{\beta} \in \mathbb{R}^{109 \times 36}$ (with row 0 containing the intercept).

2.2 Data Transformations

Variables enter the VAR in one of two forms:

- **Log-transformed** (`log`): the variable is modeled in log levels. Growth rates displayed to the user are quarter-over-quarter annualized:

$$g_t^{\text{q/q ann.}} = 400 \times \big(\ln Y_t - \ln Y_{t-1}\big). \tag{2}$$

  Year-over-year growth rates are computed as:

$$g_t^{\text{y/y}} = 100 \times \big(\ln Y_t - \ln Y_{t-4}\big). \tag{3}$$

- **Linear** (`lin`): the variable enters in levels (e.g., interest rates in percent, unemployment rate). No growth-rate conversion is applied.

2.3 Reporting Units

The display unit is chosen per variable for interpretability, independently of the estimation transformation:

- **National-accounts / activity flows** (GDP, consumption, investment, trade volumes) are shown as growth (q/q annualized or y/y).
- **Prices** (CPI/HICP/deflators) are shown as inflation.
- **Rates and ratios** (policy and market rates, unemployment, fiscal ratios) are shown as levels in percent / percentage points. Unemployment-rate uncertainty bands are floored at 0 (a rate cannot be negative); policy and market rates are *not* floored, because negative nominal rates do occur.
- **Index / FX / equity / commodity series** (the trade-weighted dollar `FXTWM`, `SP500`, house prices `HP`, oil `PZTEXP`, the non-fuel commodity index `GSNE`, and the other blocs' `NEER_EA`, `NEER_UK`, `NEER_CA`, `REER_CN`, `EQ_CN`) are reported as the *index level* (e.g. 2010 = 100) rather than as q/q growth: a trade-weighted exchange rate or equity index q/q-annualized is not an informative number, whereas its level is. These series remain `log` for estimation and are conditioned in the stored $100 \ln(\text{index})$ space; only the display is exponentiated back to the level, $\exp(\text{stored}/100)$, with a level-space uncertainty fan.

2.4 Hierarchical Minnesota Prior

The model is estimated following Giannone et al. (2015), who cast the Minnesota prior (Litterman, 1986; Doan et al., 1984) in a hierarchical framework with data-driven hyperparameter selection. The prior specification includes:

(a) The **Minnesota shrinkage** prior on B_ℓ, centered on the random-walk or white-noise prior for each variable (depending on the variable's stationarity properties). The tightness parameter λ controls the overall degree of shrinkage toward the prior mean.

(b) The **sum-of-coefficients** prior (Doan et al., 1984), which shrinks toward the belief that a unit increase in every variable's level produces a unit increase in its forecast, imposing a form of cointegration.

(c) The **single-unit-root** prior, which constrains the system to have at most one common stochastic trend.

The hyperparameters governing the tightness of each prior component are treated as additional parameters and selected by maximizing the marginal likelihood, as advocated by Giannone et al. (2015).

Posterior draws are obtained via Markov chain Monte Carlo (MCMC). The tool uses 5,000 draws with a burn-in of 2,500, retaining the posterior mean $\hat{\beta}$ and $\hat{\Sigma}$ for the deterministic conditional forecasting engine, and storing the full set of 2,500 posterior draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}_{d=1}^{D}$ for uncertainty quantification via the simulation smoother (Section 4.6).

3 Companion-Form State Space

To apply the Kalman filter, the VAR(p) in (1) is cast in companion form as a first-order state-space model (see, e.g., Durbin and Koopman, 2012, ch. 3).

3.1 State Vector

Define the np-dimensional state vector:

$$\alpha_t = \begin{pmatrix} y_t \\ y_{t-1} \\ \vdots \\ y_{t-p+1} \end{pmatrix} \in \mathbb{R}^{np}. \tag{4}$$

3.2 Transition Equation

The state evolves according to:

$$\alpha_t = T\,\alpha_{t-1} + d + R\,\eta_t, \qquad \eta_t \sim \mathcal{N}(0, \Sigma), \tag{5}$$

where the $np \times np$ transition matrix T, the np-dimensional intercept vector d, and the $np \times n$ selection matrix R are:

$$T = \begin{pmatrix} B_1' & B_2' & \cdots & B_{p-1}' & B_p' \\ I_n & 0 & \cdots & 0 & 0 \\ 0 & I_n & \cdots & 0 & 0 \\ \vdots & & \ddots & & \vdots \\ 0 & 0 & \cdots & I_n & 0 \end{pmatrix}, \quad d = \begin{pmatrix} c \\ 0 \\ \vdots \\ 0 \end{pmatrix}, \quad R = \begin{pmatrix} I_n \\ 0 \\ \vdots \\ 0 \end{pmatrix}. \tag{6}$$

Remark 1. The B_ℓ' blocks in T are transposed relative to (1) because the coefficient matrix $\hat{\beta}$ is stored as $(1 + np) \times n$, with row 0 containing the intercept c and rows $1, \ldots, np$ containing the stacked lag coefficients. Specifically, $T(i, j) = \hat{\beta}(j + 1, i)$ for $0 \leq i < n$ and $0 \leq j < np$.

3.3 Observation Equation

The observation equation extracts the contemporaneous variables:

$$y_t = Z\,\alpha_t + \varepsilon_t, \qquad \varepsilon_t \sim \mathcal{N}(0, H), \tag{7}$$

where $Z = \big(I_n \mid 0_{n \times n(p-1)}\big)$ and $H = \sigma_\varepsilon^2 I_n$ with $\sigma_\varepsilon^2 = 10^{-12}$ (a negligible measurement error included for numerical stability of the Kalman filter recursions).

3.4 State-Noise Covariance

The $np \times np$ state-noise covariance matrix is:

$$Q = R\,\Sigma\,R' = \begin{pmatrix} \Sigma & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & & \ddots & \\ 0 & 0 & \cdots & 0 \end{pmatrix}. \tag{8}$$

4 Kalman Filter and Smoother

4.1 Conditioning via Partial Observations

The key insight underlying conditional forecasting in a state-space framework, following Bańbura et al. (2015), is that conditioning on variable i at horizon h is equivalent to observing $y_{T+h,i}$ in the Kalman filter. For each forecast period $t = T + 1, \ldots, T + H$, the observation vector is partitioned into:

- **Conditioned variables:** $y_t^{(o)}$, with known target values specified by the user.
- **Free variables:** $y_t^{(m)}$, treated as missing observations (coded as `NaN`).

The Kalman filter processes only the observed subset at each time step, adapting Z_t, H_t, and the innovation v_t accordingly.

4.2 Forward Pass: Kalman Filter

Algorithm: Kalman Filter with Partial Observations. **Input:** T, Z, Q, H, d; initial state α_0, covariance P_0; partial observations $\{y_t^{(o)}\}_{t=1}^{H}$.
For $t = 1, \dots, H$:

$$\alpha_{t|t-1} = T\,\alpha_{t-1|t-1} + d \qquad \text{(state prediction)} \tag{9}$$

$$P_{t|t-1} = T\,P_{t-1|t-1}\,T' + Q \qquad \text{(covariance prediction)} \tag{10}$$

If observations exist at time t, let $Z_t^{(o)}$ be the rows of Z for observed variables, then:

$$v_t = y_t^{(o)} - Z_t^{(o)}\,\alpha_{t|t-1} \qquad \text{(innovation)} \tag{11}$$

$$F_t = Z_t^{(o)}\,P_{t|t-1}\,(Z_t^{(o)})' + H_t^{(o)} \qquad \text{(innovation variance)} \tag{12}$$

$$K_t = P_{t|t-1}\,(Z_t^{(o)})'\,F_t^{-1} \qquad \text{(Kalman gain)} \tag{13}$$

$$\alpha_{t|t} = \alpha_{t|t-1} + K_t\,v_t \qquad \text{(state update)} \tag{14}$$

$$P_{t|t} = P_{t|t-1} - K_t\,Z_t^{(o)}\,P_{t|t-1} \qquad \text{(covariance update)} \tag{15}$$

If no observations exist at time t: $\alpha_{t|t} = \alpha_{t|t-1}$ and $P_{t|t} = P_{t|t-1}$.

The matrix inversion F_t^{-1} is computed via LU decomposition with partial pivoting; singularity is flagged when the pivot falls below 10^{-15}.

4.3 Backward Pass: Durbin–Koopman Smoother

The smoother refines the filtered estimates using all future information, yielding $\alpha_{t|H}$ for all t (Durbin and Koopman, 2002, 2012).

Algorithm: Durbin–Koopman Fixed-Interval Smoother. **Input:** Filtered quantities $\{\alpha_{t|t-1}, P_{t|t-1}, v_t, Z_t^{(o)},$
Initialize $r_H = 0$. For $t = H, H-1, \dots, 1$:

If observations exist at time t:

$$L_t = T'\left(I - P_{t|t-1}\,(Z_t^{(o)})'\,F_t^{-1}\,Z_t^{(o)}\right), \tag{16}$$

$$r_{t-1} = (Z_t^{(o)})'\,F_t^{-1}\,v_t + L_t'\,r_t. \tag{17}$$

Otherwise: $r_{t-1} = T'\,r_t$.
The smoothed state is:

$$\alpha_{t|H} = \alpha_{t|t-1} + P_{t|t-1}\,r_{t-1}. \tag{18}$$

4.4 Conditional Forecast Computation

The conditional forecast is computed as follows:

1. Initialize the state from the last p historical observations: $\alpha_0 = (y'_T, y'_{T-1}, \ldots, y'_{T-p+1})'$.
2. Construct the conditioning matrix $Y^* \in \mathbb{R}^{H\times n}$, where $Y^*_{t,j}$ is the target value for variable j at horizon t if conditioned, and `NaN` otherwise.
3. Run the Kalman filter (Algorithm 4.2) and smoother (Algorithm 4.3) with partial observations given by the non-`NaN` entries of Y^*.
4. Extract the observation block (first n elements) from each smoothed state $\alpha_{t|H}$.

4.5 Delta-Based Display

To ensure that zero user deviations produce exactly the stored baseline forecast, the tool computes both an unconditional smoother pass (with all entries of Y^* set to `NaN`) and a conditional pass. The displayed forecast is:

$$\hat{y}^{\text{display}}_{T+h} = \bar{y}^{\text{baseline}}_{T+h} + \underbrace{\left(\hat{y}^{\text{cond}}_{T+h} - \hat{y}^{\text{uncond}}_{T+h}\right)}_{\text{smoother delta}}, \tag{19}$$

where $\bar{y}^{\text{baseline}}$ is the pre-computed baseline forecast (posterior mean). This construction guarantees that the displayed forecast matches the stored baseline exactly when no conditioning constraints are imposed, avoiding small numerical discrepancies between the smoother's unconditional output and the stored baseline. The unconditional smoother result is cached and reused across conditioning changes.

For log-transformed variables, the smoother operates in log-level space, and the delta is scaled to annualized growth-rate space:

$$\Delta g_h = 4 \times \Delta\ell_h, \qquad \Delta\ell_h = \ell^{\text{cond}}_h - \ell^{\text{uncond}}_h, \tag{20}$$

where ℓ_h denotes the log-level forecast at horizon h.

4.6 Uncertainty Quantification

Forecast uncertainty bands are computed via the Durbin–Koopman *simulation smoother* (Durbin and Koopman, 2002), which generates draws from the posterior predictive distribution conditional on the user's constraints.

For each posterior draw $d = 1, \ldots, D$:

1. **Forward simulation:** Draw shocks $\eta^+_t \sim \mathcal{N}(0, I_n)$ and compute a simulated path using the d-th posterior draw of the VAR coefficients:

$$\alpha^+_t = T^{(d)}\,\alpha^+_{t-1} + d^{(d)} + R\,L^{(d)}\,\eta^+_t, \tag{21}$$

where $L^{(d)}$ is the lower Cholesky factor of $\Sigma^{(d)}$. The simulated observations are $y_t^+ = Z\,\alpha_t^+$.

2. **Deviation computation:** At conditioned entries, compute $y_t^* = Y_t^{\text{target}} - y_t^+$; at free entries, set $y_t^* = \texttt{NaN}$.

3. **Smoother on deviations:** Run the Kalman smoother on $\{y_t^*\}$ with initial state $\alpha_0^* = 0$ and small initial covariance $P_0^* = 10^{-7} I_{np}$, producing adjustment paths $\hat{\alpha}_t^*$.

4. **Combine:** The conditional draw is $\tilde{\alpha}_t = \hat{\alpha}_t^* + \alpha_t^+$.

The 16th and 84th percentiles across D draws yield the reported 68% credible bands. The tool uses $D = 50$ in "Quick" mode and $D = 200$ in "Full" mode. The simulation smoother correctly integrates over both parameter uncertainty (through the posterior draws) and forecast uncertainty (through the simulated shocks).

5 Policy Counterfactuals

The counterfactual layer evaluates how macroeconomic outcomes change under an alternative policy stance. It operates *on top of* the conditional scenario described in Section 4: starting from the DK-smoothed scenario paths, the user specifies deviations in the policy rate or government spending, and the tool traces the implied effects through a DSGE model's impulse response functions before re-conditioning the full system through the BVAR.

5.1 DSGE Impulse Response Functions

Impulse response functions (IRFs) are drawn from the Macroeconomic Model Data Base (MMB; Wieland et al., 2012, 2016). The MMB provides a library of pre-solved DSGE models with standardized variable names and multiple policy rule specifications. The tool includes IRFs from 169 models under 9 monetary policy rules:

Rule	**Reference**
Taylor	Taylor (1993)
SW	Smets and Wouters (2007)
CEE	Christiano et al. (2005)
GR	Gerdesmeier & Roffia (2004)
LWW	Levin, Wieland & Williams (2003)
OW08	Orphanides & Wieland (2008)
OW13	Orphanides & Wieland (2013)
Coenen	Coenen et al. (2012)
CMR	Christiano, Motto & Rostagno (2014)

Each IRF file provides the dynamic response of key macroeconomic variables—inflation, output gap, interest rate, consumption, investment, and others—to a one-standard-deviation monetary policy shock under the specified rule. Fiscal policy IRFs are available for 60 models that include a fiscal sector.

5.2 Monetary Policy Shock Inversion

Given a user-specified interest-rate adjustment path $\Delta r = (\Delta r_1, \ldots, \Delta r_H)'$, the tool solves for the structural monetary policy shock sequence $\varepsilon = (\varepsilon_1, \ldots, \varepsilon_H)'$ via the lower-triangular Toeplitz system:

$$\underbrace{\begin{pmatrix} h_r(0) & 0 & \cdots & 0 \\ h_r(1) & h_r(0) & \cdots & 0 \\ \vdots & & \ddots & \vdots \\ h_r(H-1) & h_r(H-2) & \cdots & h_r(0) \end{pmatrix}}_{\mathcal{H}_r \in \mathbb{R}^{H\times H}} \varepsilon = \Delta r, \tag{22}$$

where $h_r(s)$ is the impulse response of the interest rate at horizon s. Since $\mathcal{H}_r$ is lower-triangular, the system is solved by forward substitution:

$$\varepsilon_t = \frac{\Delta r_t - \sum_{j=0}^{t-1} h_r(t-j)\,\varepsilon_j}{h_r(0)}, \qquad t = 0, 1, \ldots, H-1. \tag{23}$$

The implied responses of inflation and output are:

$$\Delta \pi_t^{\mathrm{IRF}} = \sum_{j=0}^{t} h_\pi(t-j)\,\varepsilon_j, \qquad \Delta y_t^{\mathrm{IRF}} = \sum_{j=0}^{t} h_y(t-j)\,\varepsilon_j. \tag{24}$$

An analogous procedure applies to fiscal policy shocks, using the government-spending IRFs (h_g, h_π^g, h_y^g, h_r^g) from the DSGE models in the MMB that include a fiscal sector.

5.3 Three Conditioning Modes

After computing the DSGE-implied deltas, the counterfactual is re-conditioned through the BVAR to enforce cross-variable consistency. Three modes govern what is held fixed versus free in the second-stage DK smoother pass:

(a) Hold-targets: All user-conditioned variables are pinned at their scenario values; the interest rate is pinned at the scenario rate plus the adjustment. Remaining variables are free and determined by the DK smoother:

$$Y_t^{\mathrm{cf}}(j) = \hat{y}_{T+t,j}^{\mathrm{cond}}, \qquad \text{for } j \in \mathcal{C}, \tag{25}$$

$$Y_t^{\mathrm{cf}}(r) = \hat{y}_{T+t,r}^{\mathrm{cond}} + \Delta r_t, \tag{26}$$

$$Y_t^{\mathrm{cf}}(j) = \texttt{NaN}, \qquad \text{otherwise.} \tag{27}$$

(b) Let-adjust: Only the policy instrument(s) are conditioned; all other variables, including inflation and output, are free and adjust endogenously through the BVAR covariance structure.

(c) IRF-consistent (default): The interest rate is pinned at the scenario rate plus the adjustment, and inflation and output are pinned at the scenario values plus the DSGE-implied deltas from (24):

$$Y_t^{\mathrm{cf}}(\pi) = \hat{y}_{T+t,\pi}^{\mathrm{cond}} + \Delta\pi_t^{\mathrm{IRF}}, \tag{28}$$

$$Y_t^{\mathrm{cf}}(y) = \hat{y}_{T+t,y}^{\mathrm{cond}} + \Delta y_t^{\mathrm{IRF}}, \tag{29}$$

$$Y_t^{\mathrm{cf}}(r) = \hat{y}_{T+t,r}^{\mathrm{cond}} + \Delta r_t. \tag{30}$$

The remaining variables are determined by the DK smoother, producing responses consistent with both the DSGE-identified monetary transmission and the BVAR covariance structure.

In all modes, the counterfactual display uses the same delta construction as (19):

$$\hat{y}_{T+h}^{\mathrm{cf,display}} = \hat{y}_{T+h}^{\mathrm{scen,display}} + f\big(\hat{y}_{T+h}^{\mathrm{cf,smooth}} - \hat{y}_{T+h}^{\mathrm{scen,smooth}}\big), \tag{31}$$

where $f(\cdot)$ applies the $\times 4$ scaling from (20) for log-transformed variables.

5.4 Optimal Policy: Barnichon–Mesters

Following Barnichon and Mesters (2023), the tool solves for the optimal monetary policy shock sequence that minimizes a quadratic loss function over deviations from target:

$$\mathcal{L} = \sum_{t=1}^{H} \Big[\lambda_\pi \big(\pi_t - \pi^*\big)^2 + \lambda_y \big(y_t - y^*\big)^2\Big], \tag{32}$$

subject to the linear mapping from shocks to outcomes given by the DSGE impulse responses.

Define the Toeplitz matrices $\mathcal{H}_\pi, \mathcal{H}_y \in \mathbb{R}^{H\times H}$ analogously to (22) using the inflation and output-gap IRFs. Stacking the weighted system:

$$\underbrace{\begin{pmatrix} \sqrt{\lambda_\pi}\,\mathcal{H}_\pi \\ \sqrt{\lambda_y}\,\mathcal{H}_y \end{pmatrix}}_{\tilde{\mathcal{H}}\in\mathbb{R}^{2H\times H}} \varepsilon = \underbrace{\begin{pmatrix} \sqrt{\lambda_\pi}\,\Delta\pi^{\mathrm{target}} \\ \sqrt{\lambda_y}\,\Delta y^{\mathrm{target}} \end{pmatrix}}_{\tilde{b}\in\mathbb{R}^{2H}}, \tag{33}$$

the optimal shock sequence is obtained via Tikhonov-regularized least squares:

$$\varepsilon^* = \big(\tilde{\mathcal{H}}'\tilde{\mathcal{H}} + \alpha I_H\big)^{-1}\tilde{\mathcal{H}}'\,\tilde{b}, \tag{34}$$

with regularization parameter $\alpha = 10^{-6}$. The implied policy rate path is $\Delta r_t^* = \sum_{j=0}^{t} h_r(t-j)\,\varepsilon_j^*$, which is then passed to the IRF-consistent conditioning mode (Section 5.3c) to obtain the full counterfactual.

The user controls the relative weight on price stability versus full employment via a slider that sets $\lambda_\pi/(\lambda_\pi + \lambda_y)$.

6 Global VAR (GVAR) and Global Commons

The single-country engine of Sections 2–5 generalizes to a multi-country setting in which a conditioning imposed on one economy spills over to all others. The tool implements a *Global VAR* (GVAR) layer in the tradition of Pesaran et al. (2004) and Dees et al. (2007), comprising the United States as a closed-economy anchor together with 23 Rest-of-World (RoW) blocs that span approximately 90% of global GDP. The decisive implementation choice is that *cross-country coupling is imposed at solve time, not at estimation time*: each bloc is an independently estimated single-country BVAR, and the spillovers enter only through the scenario *deviation*, which propagates as a linear fixed point. Version 2.0 adds a second, qualitatively distinct channel—four US-determined *global commons* (oil, the long US rate, US corporate credit, and US equity)—carried by every bloc.

6.1 Architecture: a closed anchor and 23 RoW blocs

The bloc set is $\{\mathrm{US}, \mathrm{EA}, \mathrm{JP}, \mathrm{UK}, \mathrm{CA}, \mathrm{CN}, \mathrm{CH}, \mathrm{KR}, \mathrm{MX}, \mathrm{AU}, \mathrm{TR}, \mathrm{SE}, \mathrm{PL}, \mathrm{NO}, \mathrm{DK}, \mathrm{IN}, \mathrm{BR}, \mathrm{RU}, \mathrm{ID}, \mathrm{SA}, \mathrm{AR}, \mathrm{ZA}, \mathrm{TH}, \mathrm{M}$ (24 blocs; Taiwan is excluded for want of bilateral trade data). The US is the **closed anchor**: it is *exogenous* in the Dees–PSW sense (Dees et al., 2007)—it carries *no* foreign variables, is estimated once (with $n = 37$, including the global commons of Section 6.5 as its own endogenous variables), and never iterates in the cross-country fixed point. Its influence reaches every other economy through its weight in the partners' trade-weighted aggregates.

The topology is **not hardcoded**: it self-configures from the deployed model specifications. A bloc is an active GVAR member if and only if its `spec.json` carries variables tagged `block: 'row'` *and* it has a row in the trade-weights file `weights_24bloc.json`. At load, the engine derives each bloc's GDP, price, and short-rate columns from its specification (mirroring the estimator's column-resolution order) and reads its RoW-star indices, so that bloc indices are never assumed by position. Adding an economy to the GVAR therefore requires only shipping its estimated model with the appropriate variable tags and a weights row.

6.2 RoW blocs: trade-weighted star variables

Each RoW bloc consists of its domestic VAR augmented by three trade-weighted *star* variables that summarize the foreign environment, following the weak-exogeneity device of Pesaran et al. (2004):

- `ROWDEM`: foreign demand, a trade-weighted index of partner real-GDP growth;
- `ROWINF`: foreign inflation, the analogous index on partner price levels;
- `ROWFIN`: foreign financial conditions, a trade-weighted average of partner short-term policy rates.

Let $\mathcal{W}_c \subseteq \{\text{partners of } c\}$ collect the partners that contribute to a given star at quarter t, with w_{cj} the (renormalized) trade weight of partner j in receiver c. For the activity and price stars, the partner level is read in $100\ln$ space and the star is the annualized growth index

$$\mathrm{ROWDEM}_{c,t} = 4 \sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\left(\ell^{\mathrm{gdp}}_{j,t} - \ell^{\mathrm{gdp}}_{j,t-1}\right), \qquad \mathrm{ROWINF}_{c,t} = 4 \sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\left(\ell^{\mathrm{price}}_{j,t} - \ell^{\mathrm{price}}_{j,t-1}\right), \tag{35}$$

where the scaling $4\,\Delta(100\ln\cdot)$ is the quarter-over-quarter annualized convention of (2), and the financial star is the level aggregate

$$\mathrm{ROWFIN}_{c,t} = \sum_{j\in\mathcal{W}_c} \tilde{w}_{cj}\, r_{j,t}. \tag{36}$$

The weights are *per-component* renormalized over the partners that are present and eligible at each t, $\tilde{w}_{cj} = w_{cj}\left(\sum_k w_{ck}\right)/\left(\sum_{k\in\mathcal{W}_c} w_{ck}\right)$, so that the mass of any absent or excluded partner is reabsorbed and the active weights still sum to one. Three eligibility rules mirror the estimator exactly: rate-less blocs (Switzerland, Norway) carry no own short rate and are dropped from partners' `ROWFIN` (but retained in `ROWDEM`/`ROWINF`); the hyperinflation bloc (Argentina) is dropped from partners' `ROWINF` and `ROWFIN` (retained in `ROWDEM`); and China's GDP is reported in *nominal* terms, so its activity contribution is deflated by *subtraction* in log space, $\ell^{\mathrm{gdp}}_{\mathrm{CN}} - \ell^{\mathrm{price}}_{\mathrm{CN}} = 100\ln(\mathrm{NGDP}/P)$, rather than by division.

6.3 Trade weights

The bilateral weights are constructed from IMF International Merchandise Trade Statistics (IMTS) and shipped as `weights_24bloc.json`, which lists the 24 blocs together with 23 RoW rows, each row summing to one (zero diagonal). At load, the live weight matrix W is renormalized over the *active* bloc set, so a deployment that activates a subset of the 24 candidates remains internally consistent.

6.4 Coupling at solve time, not estimation

Because each bloc is estimated independently, there is no joint, block-exogenous GVAR system; the cross-country linkage is imposed when the scenario is solved. The key observation is that, with coefficients fixed at their posterior mean $(\hat{\beta}, \hat{\Sigma})$, the conditional-forecast operator of Section 4 is *affine* in the conditioning values (the Kalman filter and Durbin–Koopman smoother are linear in the pinned data). The *baseline* forecast of each bloc is therefore data-driven and computed exactly once, via an all-`NaN` (unconditional) smoother pass; only the scenario *deviation* couples across blocs. Stacking the per-bloc RoW pin-deltas into a vector x (indexed by receiver, star variable, and horizon, of dimension $(\#\mathrm{RoW}) \cdot 3 \cdot H = 23 \cdot 3 \cdot 20 = 1380$), the deviation system is the linear fixed point

$$x = M\,x + d, \tag{37}$$

with the US frozen at its own scenario path. Here d is the exogenous forcing (the contribution of the frozen US scenario and of each bloc's own pins through its linked series), and M is the RoW-to-RoW coupling operator: receiver c's star pin (v,t) responds to partner j's free-variable level path g_j—probed once per RoW variable and horizon as an affine impulse of the smoother—weighted by w_{cj} and the appropriate difference operator ($4\,\Delta$ for the activity/price stars, the identity for the rate star). The US never appears in x; its contribution lives entirely in d.

The fixed point (37) is solved two equivalent ways:

(a) **Gauss–Seidel sweeps** (the cross-check) sum the Neumann series $x = \sum_{k\geq 0} M^k d$ by re-conditioning each RoW bloc in turn through the Durbin–Koopman smoother (Durbin and Koopman, 2002, 2012) until the maximum level change falls below tolerance.

(b) **Direct closed form**

$$x = (I - M)^{-1} d, \tag{38}$$

after which one final conditional-forecast pass per bloc reconstructs the full set of free variables consistent with the converged pin-deltas.

At the full 24-bloc configuration the empirical spectral radius is $\rho(M) \approx 0.72 < 1$, so the Neumann series converges and $(I - M)$ is nonsingular; the two solvers agree to $\sim 10^{-9}$.

A practical consequence drives the deployment. The operator M depends on the estimated models and on the conditioning *structure* (which RoW cells are pinned) but *not* on the pinned *values*, so it can be assembled once and reused for every value change during interactive dragging. Its dense inverse is, however, expensive: at dimension 1380 the naive in-browser inversion would take on the order of tens of minutes (roughly 19 minutes measured), making a cold build infeasible even in a Web Worker. The tool therefore **ships a precomputed coupling artifact** (`coupling_default.json`) containing M, $(I-M)^{-1}$, and $\rho(M)$ for the default (empty) pin structure. This artifact is mandatory, not an optimization: hydrating it reduces the first interactive solve from ~ 19 minutes to sub-second. A model-signature gate guards against using a stale inverse after any re-estimation; a genuinely novel RoW-bloc pin structure misses the cache and falls back to a live build.

6.5 Global commons: US-determined common shocks

The headline addition of version 2.0 is a set of four *global common* variables that are determined by the United States but carried by *every* bloc:

- `GLB_OIL` — the spot oil price (WTI);
- `GLB_USLR` — the US 10-year Treasury yield;
- `GLB_USBAA` — the US Baa corporate-bond yield (a credit-spread proxy);
- `GLB_USEQ` — US equity prices.

In the US anchor these enter as ordinary *endogenous* variables; in each RoW bloc the same four identifiers (matched one-to-one by *name*, never by position) are carried as weakly-exogenous commons that the solver *pins*.

The pinning is by *deviation*, not by level. The US is solved first and frozen; each RoW bloc c then sets each global column to its own baseline plus the US scenario deviation of the matching global,

$$\widehat{\mathrm{GLB}}_{c,t} = \mathrm{G\bar{L}B}_{c,t} + \left(\mathrm{GLB}^{\mathrm{scen}}_{\mathrm{US},t} - \mathrm{G\bar{L}B}_{\mathrm{US},t}\right), \tag{39}$$

where $\bar{(\cdot)}$ denotes the bloc's own unconditional baseline. This is a load-bearing design choice. Each RoW bloc estimated its *own* baseline path for the globals as weakly-exogenous regressors, and those baseline levels differ from the US's by tens of $100\ln$ units (for example, of order $+17$, $+70$, and $+28$ for the euro area, Japan, and China on oil). Pinning the US *level* would inject that entire baseline gap as a spurious deviation the instant a bloc is selected, even under a zero scenario. Pinning the US *deviation* on top of each bloc's own baseline instead yields exactly zero spillover at a zero scenario and an *identical* common shock under conditioning—precisely mirroring the star convention $\mathrm{R\bar{o}W}$ + partner-delta of (35)–(36). A US oil pin therefore propagates the same oil deviation to all 23 RoW blocs (verified to within 10^{-9}), while each bloc's domestic variables respond through its *own* estimated coefficients.

Mechanically, the global pins are a US→RoW *exogenous* forcing (the US is frozen), so they enter the per-bloc forcing d of (37) and *not* the RoW-to-RoW coupling M: they are held fixed across the per-period probes that assemble M, and so cancel out of it identically. Consequently the precomputed inverse $(I - M)^{-1}$ is unaffected by the globals and remains valid without regeneration. Indirect effects—a common shock moving a bloc's macroeconomy, which moves its stars, which spill to other blocs—flow through the existing star coupling M and are not double-counted. A bloc that carries no globals reduces the pin to a no-op, leaving the legacy globals-free solve byte-identical.

On the user interface the globals are *conditionable* on the US screen (where they are endogenous) and *read-only*, carried over from the US path, on every other bloc.

6.6 Relation to the fiscal and counterfactual layers

The GVAR layer composes with the single-country machinery rather than replacing it. The displayed scenario line for an active GVAR bloc is the cross-country propagated path of this section; for a standalone (non-GVAR) economy it is the single-country conditional forecast of Section 4. The US derived fiscal block (the debt-to-GDP accounting identity, whose empirical motivation is the fiscal reaction function of Bohn, 1998) and the DSGE policy counterfactual—including the Barnichon and Mesters (2023) optimal-policy problem—continue to operate on the US bloc's propagated path. Conditional forecasts throughout use the large-cross-section state-space method of Bańbura et al. (2015) with the hierarchical prior of Giannone et al. (2015), applied independently to each bloc and stitched together only through the linear deviation system (37).

7 Fiscal Blocks: Effective-Rate Drivers and Debt Accounting

Six of the economies in the tool—the United States, the Euro Area, Japan, the United Kingdom, Canada, and China—carry a dedicated *fiscal block*. The block follows the stock-flow-consistent division of labour shared by the U.S. Congressional Budget Office's conditional-BVAR budget mechanics and the IMF Financial Programming (FPP) / debt-sustainability frameworks: the BVAR is used as a *macro-fiscal scenario generator* that jointly forecasts the macroeconomy together with a small number of fiscal *flow drivers*, while the public-debt stock and the headline reporting variables (primary and overall balance, interest, debt-to-GDP, gross financing needs) are *derived outside the VAR* by the government budget-constraint identity. No reduced-form equation is ever allowed to forecast the debt stock freely, because an unconstrained debt VAR is explosive: the debt-dynamics identity has a near-unit autoregressive root $(1 + i)/(1 + g^n)$, and small parameter errors compound into divergent debt paths. Imposing the identity by construction is exactly the discipline that the CBO and FPP/DSA frameworks demand, and it is the structure implemented in `app/src/lib/compute/fiscalAccounting.ts`.

7.1 Estimated Fiscal Drivers

The BVAR jointly estimates the macro block of Section 9 together with three fiscal flow drivers, all entering as linear (`lin`) variables:

- **Revenue** (`FREV_GDP`): general-government receipts as a percentage of GDP, an annual ratio.
- **Primary expenditure** (`FPEXP_GDP`): non-interest outlays as a percentage of GDP, an annual ratio.
- **Effective interest rate on debt** (`EFFI`): the weighted-average rate paid on the outstanding stock, in percent per annum.

The use of the effective rate is the central modelling choice of the v2.0 fiscal block. The earlier parametrisation carried Δ(debt/GDP) directly as a driver (the `DDEBT_GDP` variable of Table 2) and backed the debt level out by cumulating it; this conflated the automatic debt dynamics with the discretionary primary balance and gave the user no clean handle on the debt-service channel. In v2.0, the Δ(debt/GDP) driver is replaced by the effective rate `EFFI`, and debt is re-derived from the budget identity (Section 7.2). The effective rate is the *average* rate on the existing debt portfolio—it lags market yields because the stock turns over only as maturing securities are refinanced—and is therefore distinct from, and estimated separately from, the policy rate and the market yields already in the macro block.

Because revenue, primary expenditure, and the effective rate are ordinary BVAR variables, they inherit the full estimation machinery of Sections 2–4: the hierarchical Minnesota prior (Giannone et al., 2015), the COVID-as-missing data-augmentation Gibbs treatment, the companion-form Durbin–Koopman conditional smoother (Durbin and Koopman, 2002, 2012), and the Bańbura et al. (2015) conditioning-via-partial-observations construction. In the five Global-VAR blocs (EA, JP, UK, CA, CN) the fiscal drivers are part of the domestic block and propagate to the other economies through the shared Rest-of-World stars, exactly as the macro variables do (Pesaran et al., 2004; Dees et al., 2007).

Each fiscal bloc ships a `fiscal_meta.json` file that pins the column layout used by the accounting recursion. It carries the integer indices $\texttt{idx} = \{\texttt{gdp}, \texttt{deflator}, \texttt{revenue}, \texttt{primaryExp}, \texttt{effRate}\}$ into the level-space forecast matrix—the first two locate the real-GDP and deflator log-levels that supply nominal growth, the last three locate the fiscal drivers—together with the seed debt ratio d_0 (e.g. the U.S. block has $d_0 = 122.57$ and $\texttt{idx.effRate} = 35$; the U.K. block has $d_0 = 96.24$). The presence of `idx.effRate` is the switch that selects the effective-rate recursion; a legacy block lacking it would fall back to the old Δ-debt construction.

7.2 The Budget-Identity Recursion

Let d_t denote the debt-to-GDP ratio (in percent), i_t^{eff} the effective rate (percent per annum), and g_t^n nominal GDP growth (percent per quarter). All flows below are annual ratios in percent of GDP. Nominal growth is read directly off the level-space forecast as the sum of the real-GDP and deflator log-differences, in the $100 \cdot \ln$ space of Section 2.3:

$$g_t^n \;=\; \big(\ell_t^{\text{gdp}} - \ell_{t-1}^{\text{gdp}}\big) \;+\; \big(\ell_t^{\text{defl}} - \ell_{t-1}^{\text{defl}}\big), \tag{40}$$

where ℓ is the stored $100 \cdot \ln$ level and the right-hand side is already in percent per quarter (it is not annualized by the factor of four used for display in (2)). The primary balance is the revenue–expenditure gap (positive = surplus),

$$pb_t \;=\; rev_t \;-\; primexp_t, \tag{41}$$

interest (debt service) is the effective rate applied to *lagged* debt,

$$int_t \;=\; \frac{i_t^{\text{eff}}\, d_{t-1}}{100}, \tag{42}$$

and the overall balance and deficit follow as

$$ob_t \;=\; pb_t - int_t, \qquad def_t \;=\; -\,ob_t \;=\; int_t - pb_t. \tag{43}$$

The debt stock is then rolled forward one quarter by the FPP/DSA debt-dynamics equation,

$$d_t \;=\; d_{t-1}\,\frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n/100} \;-\; \frac{pb_t}{4} \;+\; \frac{sfa_t}{4}, \qquad d_{-1} \equiv d_0, \tag{44}$$

seeded at d_0. The rate enters quarterized as $i_t^{\text{eff}}/400$ and growth as $g_t^n/100$, while the annual flow ratios pb_t and sfa_t are scaled by $1/4$ so that one quarter's worth of the flow accrues per step; sfa_t is the stock-flow adjustment ("other means of financing," valuation effects, below-the-line operations), defaulting to zero. Gross financing needs add amortization to the deficit,

$$gfn_t \;=\; def_t \;+\; amort_t. \tag{45}$$

The recursion in (41)–(45) is the function `computeFiscalBlock` (with the lever-augmented variant in `runFiscalRecursion`); the exact quarterly convention is pinned by a historical back-cast test that reconstructs observed debt-to-GDP from the realized drivers.

Snowball decomposition. Subtracting d_{t-1} from (44) isolates the *automatic* debt dynamics from the discretionary primary balance. The change in the debt ratio is

$$\Delta d_t \;=\; \underbrace{\left(\frac{1+i_t^{\text{eff}}/400}{1+g_t^n/100}-1\right)d_{t-1}}_{\text{snowball (interest–growth)}} \;-\; \frac{pb_t}{4} \;+\; \frac{sfa_t}{4} \;\approx\; \frac{i_t^{\text{eff}}-g_t^n}{1+g_t^n}\,d_{t-1} \;-\; pb_t \;+\; sfa_t, \tag{46}$$

where the right-hand approximation is the familiar Fiscal-Monitor $(i-g)$ form (here in annualized flow units). The snowball term is positive—debt rises mechanically—whenever the effective rate exceeds nominal growth; a primary surplus and a favourable growth-interest differential are the two forces that stabilize the ratio. The tool returns the per-quarter snowball contribution $\big((1+i_t^{\text{eff}}/400)/(1+g_t^n/100)-1\big)d_{t-1}$ alongside the derived balances so the user can read off the source of any debt drift.

Remark 2. Because the debt stock is *not* a BVAR variable, the estimated effective rate $\hat{i}_t^{\text{eff}}$ and primary balance $\widehat{pb}_t$ are debt-agnostic baselines: the recursion uses *lagged* debt d_{t-1} throughout, so it stays explicit (no within-period fixed point) and the active levers of Section 7.3 layer on without double-counting. This is the design that lets a single forward sweep produce a consistent debt path.

7.3 Active Levers

The accounting layer exposes three user-editable levers with literature-based defaults, governed by a master toggle (`feedbackOn`, default on). They modify the estimated baselines in terms of the lagged debt gap $d_{t-1}-d^*$, where the anchor d^* defaults to d_0.

Bohn fiscal-reaction function (γ). Following Bohn (1998), the primary balance is allowed to respond to the lagged debt gap, so that fiscal policy consolidates as debt rises above its anchor:

$$pb_t \;=\; \big(rev_t - primexp_t\big) \;+\; \gamma\,\big(d_{t-1}-d^*\big), \qquad \gamma \geq 0. \tag{47}$$

The default $\gamma = 0.03$ (three basis points of primary surplus per percentage point of excess debt) is the canonical Bohn (1998) estimate and the threshold above which a no-Ponzi debt process is mean-reverting. Setting $\gamma = 0$ disables the reaction.

Risk-premium channel (β). The effective rate is allowed to rise with debt above the anchor, capturing a sovereign risk premium:

$$i_t^{\text{eff}} \;=\; \hat{\imath}_t^{\text{eff}} \;+\; \beta \max\!\big(0,\; d_{t-1} - d^*\big), \qquad \beta \ge 0, \tag{48}$$

where the premium bites only *above* the anchor (the $\max(0, \cdot)$ rectifier). The default is $\beta = 0$: the magnitude of the debt-premium elasticity is empirically contested, so the lever is shipped inactive and left to the user. When active, β feeds back into both the interest bill (42) and the snowball term (46), producing the nonlinear debt-spiral dynamics that motivate the FPP/DSA realism literature.

Stock-flow-adjustment closure. The SFA term sfa_t in (44) is a soft-identity residual. It may be left at zero (pure flow accounting), set to the historically estimated average, or used as a scenario lever to reconcile a target debt path with the flow drivers.

Both feedback channels act on the lagged gap and through the lagged debt already entering (42)–(44), so they are added cleanly to the debt-agnostic baselines without re-estimation and without double-counting.

7.4 Debt-Targeting Inversion

The recursion can also be run backwards. Given a desired debt-to-GDP path $\{d_t^{\text{target}}\}$, the effective rate, nominal growth, and SFA, the annual primary balance required at each quarter is obtained by solving (44) for pb_t along the target trajectory:

$$pb_t \;=\; 4\left[d_{t-1}^{\text{target}} \,\frac{1 + i_t^{\text{eff}}/400}{1 + g_t^n/100} \;-\; d_t^{\text{target}} \right] \;+\; sfa_t, \tag{49}$$

with $d_{-1}^{\text{target}} \equiv d_0$. This is the IMF DDT/DSA "required primary balance / adjustment to the debt anchor" (`requiredPrimaryBalance`). The caller then conditions the BVAR on this primary-balance path through the linear constraint $rev_t - primexp_t = pb_t$, propagated by the DK smoother of Section 4. Because the effective rate and growth themselves respond to the primary balance through the BVAR covariance structure, an exact hit is obtained by iterating invert $\to$ condition $\to$ re-read (i, g) $\to$ re-invert; one or two passes converge.

7.5 Display Layout

The estimated and derived fiscal objects are presented in two different chart rows, reflecting their different epistemic status. The *effective rate* `EFFI` sits in row 2 as an estimated driver, shown as a level in percent per annum (`% p.a.`), next to the other rates and ratios of Section 2.3. Row 3 holds the three *derived* reporting variables produced by the accounting recursion, in the order

(i) **Primary Balance** (pb_t, % of GDP), which is itself conditionable: the user can drag a primary-balance path and the tool maps it back to the $rev - primexp$ linear combination;

(ii) Δ **Debt-to-GDP** (Δd_t, percentage points per quarter), the per-quarter debt change of (46), which is not directly conditionable because it is a pure identity output;

(iii) **Debt-to-GDP** (d_t, % of GDP), the rolled-forward stock, which can be targeted via the inversion of Section 7.4.

Separating the estimated effective-rate driver (row 2) from the derived balances and debt (row 3) makes the stock-flow-consistent structure visible on screen: what the data and the prior speak to is shown as an input, and what the budget identity implies is shown as an output.

7.6 Data Treatment: the U.K. Index-Linked-Gilt Spike

Effective-rate series constructed from accrued interest can contain mechanical, non-recurring spikes that are not informative about the underlying refinancing cost. In the United Kingdom, the large stock of *index-linked gilts* produced an effective-rate spike in 2008–09: the inflation uplift on the principal of index-linked debt is recorded as accrued interest, so the 2008 inflation spike (and its subsequent unwinding) translated into an outsized swing in the measured `EFFI` that would otherwise contaminate the BVAR's autoregressive dynamics. These quarters are handled with the same machinery used for the COVID-19 observations: the affected effective-rate entries are set to missing (`NaN`) and imputed by the Kalman filter inside the data-augmentation Gibbs sampler, rather than being allowed to enter the likelihood as if they were ordinary observations. This keeps the estimated effective-rate process—and hence the derived interest bill and debt path—free of a one-off accounting artefact while preserving the genuine information in the surrounding quarters.

8 Long-Run Anchoring: Prior vs Entropic Tilt

The forecasts of Section 4 inherit their long-run behaviour entirely from the estimated VAR. Because the real-activity and price blocks are estimated on $100 \ln$ levels of $I(1)$ series, the companion matrix (6) carries roots on or near the unit circle, and the implied long-horizon central path can drift away from any value the user regards as a credible steady state (e.g. a 2% inflation objective, a neutral policy rate, or a balanced great ratio). The tool exposes a *long-run anchor* control that lets the user pin the long-run forecast of a chosen linear combination $h'y$ to a target. We support two conceptually distinct mechanisms, summarized in Table 1: an *in-prior* long-run prior applied *before* estimation, and an *after-estimation* entropic tilt of the predictive density. They answer different questions, and the distinction is methodologically important.

8.1 In-Prior Long-Run Prior (PLR)

The first mechanism is the "prior for the long run" (PLR) of Giannone et al. (2019), which augments the hierarchical Minnesota prior of Section 2.4 with additional dummy observations that shrink a small set of linear combinations Hy toward stationarity (no-cointegration / random-walk behaviour). Let $H \in \mathbb{R}^{q\times n}$ collect q long-run combinations and let y_0 be the prior pre-sample mean. Following the authors' construction, H is rotated into a full-rank basis $\bar{H} = \big(H'; \operatorname{null}(H)'\big)'$ with inverse $\bar{H}^{-1}$, and for each active long-run row j a single dummy observation

$$y_j^d = \frac{(y_0 \bar{H}')_j}{\phi_j} \left(\bar{H}^{-1}\right)'_{\cdot j}, \qquad x_j^d = \Big(0,\ \tfrac{(y_0 \bar{H}')_j}{\phi_j}\, \mathbf{1}_p \otimes \left(\bar{H}^{-1}\right)'_{\cdot j}\Big), \tag{50}$$

is appended *to the regression system* (both the Gibbs draw and the marginal-likelihood normalizing constant T_d), with ϕ_j controlling the tightness of the j-th long-run restriction. Crucially, (50) combines the *equations* through the columns of $\bar{H}^{-1}$; combining the regressors instead produces a dummy whose residual is identically zero at the sum-of-coefficients mode and hence carries no tightness information. The tightness hyperparameters ϕ_j are held fixed so that the Metropolis–Hastings search over the Minnesota λ and the sum-of-coefficients μ remains two-dimensional.

The PLR is a *proper Bayesian prior*: it modifies the posterior and requires re-estimation. It is therefore computed offline as part of the data-export pipeline, not in the browser.

Remark 3 (What the PLR actually does in the US sample). A US head-to-head ($n = 36$, $p = 3$, COVID treated as missing per Section 4.6, $q = 13$ long-run combinations comprising six great ratios, three nominal anchors, and four rate/credit spreads) shows that the PLR is a *separate dummy block* rather than a change to the Minnesota tightness: the maximizing λ is essentially unchanged (0.563 baseline vs 0.562 with the PLR at either $\phi = 1.0$ or $\phi = 0.1$). Because the sum-of-coefficients-style dummy in (50) shrinks toward a *random walk* (no cointegration), and the US sample treats most of these ratios as genuinely $I(1)$, the PLR does *not* pull the long-run central forecast toward a historical mean: the mean absolute drift of the H=40 forecast from a recent historical anchor moves from 8.49 (baseline) to 8.41 at $\phi = 1.0$ (-1%), and is marginally *worse* at $\phi = 0.1$. The visible effect is on the *bands*, not the path: the GDP-growth 68% band tightens from 4.24 to 3.95 percentage points ($\approx 7\%$) at $\phi = 1.0$. The PLR is thus the appropriate instrument for principled band tightening and a soft no-cointegration restriction; it is *not* a device for pinning a long-run combination to a chosen number.

8.2 After-Estimation Entropic Tilt

The second mechanism is the user-facing one, exposed in the interface as the **Long-run priors** panel (Section 10). It does *not* re-estimate the model. Instead it takes the *existing* posterior draws $\{(\hat{\beta}^{(d)}, \hat{\Sigma}^{(d)})\}_{d=1}^{D}$ of Section 4.6 and applies an exponential (entropic) tilt — a relative-entropy reweighting in the spirit of Robertson et al. (2005) — so that the *weighted* mean of each long-run moment matches the user's target. The result is a vector of per-draw importance weights that the band loop applies through weighted quantiles; the underlying draws are never re-solved.

The long-run moment. For draw d the relevant long-run quantity is taken to be the *long-horizon mean forecast*, not the algebraic steady state:

$$g^{(d)} = h' \, \mathbb{E}\big[y_{T+H} \mid \hat{\beta}^{(d)}\big], \tag{51}$$

obtained by iterating the deterministic companion recursion (5) forward H steps with the innovations switched off, exactly as in the mean iteration underlying (21). The default horizon is $H = 40$ (≈ 10 years quarterly), matching the convention of Giannone et al. (2019). We deliberately do *not* use the closed-form steady state $y^\star = (I - \sum_\ell B_\ell)^{-1}c$, $g = h'y^\star$. For the near-unit-root level VARs the matrix $(I - \sum_\ell B_\ell)$ is near-singular, so $y^\star$ is a meaningless quantity of order 10^5: e.g. for the US GDPH − CH log spread the steady-state evaluation returns $\approx -2.1 \times 10^4$, whereas the H=40 forecast moment is the finite $\approx +35$ that matches the chart. The steady-state evaluation is retained only as an opt-in for genuinely stationary combinations.

For a single user variable the moment is stated in the chart's own display units (Section 2.3): for a `lin` variable (rates, unemployment) the target is the forecast *level* $e_j'\mathbb{E}[y_{T+H}]$; for a `log` variable it is the annualized growth $4(\mathbb{E}[y_{T+H,j}] - \mathbb{E}[y_{T+H-1,j}])$, consistent with (20), so the user enters the number they read off the axis.

KL-minimizing weights. Collecting q restrictions with targets $\bar{g}_r$ and tightness scales s_r, define the centered, scaled moments $\tilde{g}_r^{(d)} = (g_r^{(d)} - \bar{g}_r)/s_r$. We seek weights $w_d \propto \exp\big(\gamma'\tilde{g}^{(d)}\big)$ that satisfy the moment conditions $\sum_d w_d\, g_r^{(d)} = \bar{g}_r$ for all r while minimizing the Kullback–Leibler divergence (Kullback and Leibler, 1951) from the uniform weights $1/D$. This is the classical exponential-tilting / minimum-relative-entropy projection: the tilted measure is the I-projection of the empirical predictive measure onto the affine set of distributions satisfying the long-run moment constraints. The Lagrange dual is the smooth, convex objective

$$F(\gamma) = \log\left(\frac{1}{D}\sum_{d=1}^{D}\exp\big(\gamma'\,\tilde{g}^{(d)}\big)\right), \qquad \nabla F(\gamma) = \mathbb{E}_w\big[\tilde{g}\big], \qquad \nabla^2 F(\gamma) = \mathrm{Cov}_w\big(\tilde{g}\big), \tag{52}$$

whose stationary point $\nabla F(\gamma^\star) = 0$ enforces the constraints. We solve (52) by Newton iteration with a backtracking (Armijo) line search and a 10^{-10} ridge on the Hessian, computing F via a numerically stable log-mean-exp; the weights match the target mean exactly regardless of the scales s_r, which serve only to balance restrictions of different magnitudes (each set to the moment's own posterior standard deviation). The normalized weights w_d are then applied as weighted quantiles to *both* the baseline fan (the precomputed band of Section 4.6 is replaced by the in-browser tilted band) and the conditional/scenario fan, so the entire predictive density is pulled toward the anchor.

Effective sample size and degeneracy. The tilt is a pseudo-posterior reweighting, and its reliability is governed by the effective sample size

$$\mathrm{ESS} = \frac{\big(\sum_d w_d\big)^2}{\sum_d w_d^2} = \frac{1}{\sum_d w_d^2} \in [1, D], \tag{53}$$

the harmonic measure of how many draws carry non-negligible mass after reweighting. As the target is pushed farther from the prior mean $D^{-1}\sum_d g^{(d)}$, the weights concentrate and ESS falls. The interface surfaces ESS/D as a color-coded badge (green above 40%, amber between 10% and 40%, red below 10%) so the user can see when an anchor is straining the support of the draws.

Remark 4 (Per-constraint vs joint tilt). In the US comparison of Remark 3, with $D = 500$ draws a *per-constraint* tilt — anchoring one combination at a time — hits every target exactly with a healthy effective sample size (median ESS $\approx$ 75.9% of D). The *joint* 13-constraint tilt, by contrast, is degenerate: ESS $\to 1$, because the joint anchor target lies outside the 13-dimensional convex hull of only 500 draws even though each marginal target lies inside its own one-dimensional hull. The practical consequence, reflected in the interface, is that the entropic tilt is well-behaved for one or a few long-run restrictions but should not be over-constrained: the panel defaults to a single anchor and warns when several are enabled simultaneously.

8.3 When to Use Which

The two mechanisms are complementary. The in-prior PLR is the gold standard — a proper offline posterior — but its data-decides property means that, at the tightnesses examined, it leaves the US central long-run forecast essentially unchanged and acts mainly on the bands. The entropic tilt is in-browser, requires no re-estimation, and *forces* the weighted long-run forecast onto an explicit target, but it is a reweighting of a finite draw cloud and degrades (falling ESS) if over-constrained. As a rule of thumb: to *pin* a long-run combination to a number, use the entropic tilt (or, offline, a tighter targeted prior); for principled band tightening and a soft no-cointegration restriction, use the PLR.

Table 1: Long-run anchoring: in-prior PLR versus after-estimation entropic tilt.

	In-prior PLR	**Entropic tilt**
Source	Giannone et al. (2019)	Robertson et al. (2005)
Acts on	prior (dummy obs.)	posterior draws (weights)
Re-estimation	yes (offline)	no (in-browser)
Object	proper posterior	pseudo-posterior
Moment	Hy stationarity	$h'\mathbb{E}[y_{T+H}]$
Main effect	tightens bands	shifts central path
Failure mode	none (proper)	low ESS if over-constrained

9 Economy Coverage, Data Sources, and Estimation

The tool ships estimated BVARs for roughly fifty economies. Twenty-four of them are integrated into a single *Global VAR* (GVAR) system (Pesaran et al., 2004; Dees et al., 2007) covering approximately 90% of world GDP; the remaining blocs are standalone single-country models that share the same estimation machinery but are not (yet) wired into the cross-country fixed point. This section documents the roster, the underlying data, the COVID-as-missing estimator, the common-vintage harmonization, and the construction of the predictive bands. Throughout, variables enter the

VAR in the $100\ln(\cdot)$ space for `log` series and in levels for `lin` series (Section 2.3); annualized quarter-over-quarter growth and inflation are reported as $4 \times \Delta(100\ln)$, consistent with (2) and (20).

9.1 The GVAR System: 24 Integrated Blocs

The GVAR comprises the United States as the closed-economy anchor plus twenty-three Rest-of-World (RoW) blocs:

> US (anchor); EA, JP, UK, CA, CN (the five core majors); and CH, KR, MX, AU, TR, SE, PL, NO, DK, IN, BR, RU, ID, SA, AR, ZA, TH, MY.

Each non-US bloc is an independently estimated single-country BVAR in which the foreign sector is summarized not by a raw foreign-variable block but by three trade-weighted RoW "star" variables, following the GVAR tradition of Pesaran et al. (2004) and Dees et al. (2007):

- `ROWDEM` — foreign demand: the trade-weighted average of partners' real-GDP growth;
- `ROWINF` — foreign inflation: the trade-weighted average of partners' price inflation;
- `ROWFIN` — foreign financial conditions: the trade-weighted average of partners' short rates.

For bloc c with partner set $\mathcal{P}_c$ and weights w_{cj}, the demand and inflation stars are built in annualized growth space from the partners' stored $100\ln$ levels,

$$\mathrm{ROWDEM}_{c,t} = 4 \times \frac{\sum_{j\in\mathcal{P}_c} w_{cj}\left(x^g_{j,t} - x^g_{j,t-1}\right)}{\sum_{j\in\mathcal{P}_c} w_{cj}\,\mathbb{1}\{x^g_{j,t}, x^g_{j,t-1}\text{ observed}\}}, \tag{54}$$

where $x^g_{j,t}$ is partner j's real-activity level (analogously for `ROWINF` from prices, and `ROWFIN` from the level of the short rate). Weights are *renormalized per component* over the partners actually observed in each quarter, so a partner that is missing a given series simply drops out of the corresponding star and the remaining weights are rescaled. Two implementation details matter for the econometrician: China's reported GDP is nominal, so its real-activity level is formed by subtracting the price level in $100\ln$ space, $x^g_{\mathrm{CN}} = x^{\mathrm{NGDP}}_{\mathrm{CN}} - x^P_{\mathrm{CN}}$ (a log-space deflation); and Argentina's 2023–24 hyperinflation episode is excluded from the `ROWINF` and `ROWFIN` aggregates of its partners (its genuine real demand is retained in `ROWDEM`), with renormalization absorbing the dropped weight.

Trade weights. Bilateral weights w_{cj} are constructed from goods-trade flows in the IMF International Trade in Goods statistics (IMTS, the renamed bilateral DOTS), queried from the SDMX 2.1 endpoint `api.imf.org` over the 2021–2023 average of exports (`XG_FOB_USD`) plus imports (`MG_CIF_USD`). The Euro Area is treated as a single counterpart (IMTS aggregate code `G163`). The weight matrix is row-normalized over the 24 blocs with a zero diagonal; the US carries no RoW block and therefore no row.

Rate-optional blocs. The short rate is *optional*: a GDP series and a price series are required of every bloc, but two blocs (Switzerland and Norway) carry no domestic short rate. Their rate column is `NaN` throughout, they are excluded from their partners' `ROWFIN` aggregate, and the `ROWFIN` weights of any bloc having them as a partner are renormalized accordingly. They still contribute to `ROWDEM` and `ROWINF`.

Cross-country coupling. As in Section 9's original treatment, the baseline forecast of each bloc is computed independently; cross-country coupling enters only through the scenario layer. A shock conditioned in one bloc perturbs its star variables, which propagate to its trade partners, whose own stars then feed back. Stacking the per-bloc deviations Δ yields the linear fixed point

$$\Delta = M\,\Delta + d, \tag{55}$$

where M encodes the trade-weighted star couplings and d collects the user's direct conditioning forcing plus the US-determined global commons (oil, the trade-weighted dollar, etc.), which are pinned one-for-one into each RoW bloc. The system is solved at display time both by a direct $(I - M)^{-1}d$ evaluation and by Gauss–Seidel sweeps (summing the Neumann series $\sum_k M^k d$); the two agree to $\sim 5 \times 10^{-11}$. The coupling is mild, with spectral radius $\rho(M) \approx 0.59$, so roughly ten sweeps suffice. Because the 24-bloc $(I - M)^{-1}$ inverse is expensive in-browser, it is precomputed and shipped as a coupling cache.

9.2 Standalone Single-Country Blocs

Beyond the 24 GVAR blocs, the tool ships additional single-country BVARs estimated by the same procedure but not coupled into (55). Together with the GVAR members these complete coverage of all 38 OECD members and add several large non-OECD emerging markets. The non-GVAR remainder comprises the OECD economies AT, BE, CL, CO, CR, CZ, EE, FI, FR, DE, GR, HU, IS, IE, IL, IT, LV, LT, LU, NL, NZ, PT, SK, SI, ES, plus the non-OECD market TW (Taiwan). The country selector groups the roster into *Core* (the six majors), *OECD*, and *EM* tiers; a bloc is GVAR-linked if and only if it has a trade-weights row and its specification carries the three `ROW*` star variables.

9.3 Data Sources

No single provider supplies current quarterly real GDP for the full roster, so the data are assembled bloc-by-bloc:

- **FRED.** The US block ($n = 36$, Table 2) and many other series are pulled from the St. Louis Fed's FRED. The China bloc is built entirely from verified real FRED series, with quarterly real activity anchored by industrial production (`PRCNTO01CNQ661S`) since clean quarterly *real* GDP is not available on FRED for China (only annual constant-price GDP and nominal quarterly GDP that ends early).

- **Eurostat.** The IMF's new thematic SDMX API exposes no per-country quarterly chain-volume GDP (its national-accounts flow carries current-price values only, for a limited country set), so the 22 EU/EFTA OECD economies are sourced from Eurostat (`namq_10_gdp` chain-linked volumes, `prc_hicp_midx` HICP, `une_rt_q` unemployment, `irt_*` interest rates, `ert_eff_ic_q` NEER), which is current to 2025Q4–2026Q1.

- **National banks and OECD QNA.** The non-EU OECD members (e.g. Chile, Korea, Australia, New Zealand, Mexico, Türkiye) draw real GDP and national accounts from the OECD Quarterly National Accounts, short rates from national-bank series (e.g. RBA/RBNZ), and CPI, unemployment, and NEER from the IMF flows.

Real versus nominal GDP is handled per bloc: where a chain-volume real series exists it is used directly; China's nominal GDP is deflated in log space as above; and thin or volatile samples are estimated with deliberately wide, honest predictive bands rather than spuriously tight ones.

9.4 COVID-as-Missing Estimation

The 2020 pandemic quarters are an extreme outlier that, left untreated, would contaminate the autoregressive estimates and the innovation covariance. Rather than scaling them out, the tool treats the pandemic window as *missing data* and imputes it within the estimation itself, via a data-augmentation Gibbs sampler that nests a Durbin–Koopman simulation smoother (Durbin and Koopman, 2002, 2012). Each Gibbs sweep alternates:

(i) a Durbin–Koopman simulation-smoother draw of the missing pandemic rows given the current coefficients, treating those rows as `NaN` observations in the companion-form state space of Section 3;

(ii) a Metropolis–Hastings update of the hierarchical-prior hyperparameters on the *completed-data* marginal likelihood, so the overall tightness λ and the sum-of-coefficients parameter μ (Giannone et al., 2015) are selected by empirical Bayes without any fixed-tightness restriction; and

(iii) a draw of the VAR coefficients (β, Σ) from the completed data.

The pandemic history is thus *preserved* (the surrounding quarters remain in the sample and inform the imputation) while the outlier itself does not drive the dynamics.

Window selection. The COVID window self-calibrates to each bloc's pandemic trough rather than being hard-coded. It is the *argmin* of the quarter-on-quarter GDP change *restricted to the window 2019Q4–2021Q1*:

$$k^\star = \underset{t \in \{2019\text{Q4},\ldots,2021\text{Q1}\}}{\arg\min} \left(x_t^g - x_{t-1}^g\right), \qquad \mathcal{W} = \{\, k^\star - 1,\ k^\star,\ k^\star + 1,\ k^\star + 2 \,\}, \tag{56}$$

and the four rows of $\mathcal{W}$ (intersected with the sample) are NaN'd across *all* variables before estimation. Restricting the search to 2019Q4–2021Q1 is deliberate: an unrestricted global argmin misfires for economies whose deepest contraction is a different crisis (for example Korea's 2008Q4 Global-Financial-Crisis drop exceeds its 2020Q2 pandemic drop).

9.5 Common-Vintage Harmonization to 2025Q4

So that the first forecast (nowcast) quarter is uniform across the tool, every bloc's domestic series is extended to a common 2025Q4 endpoint, making 2026Q1 the first projected quarter everywhere. For blocs whose quarterly data end short of 2025Q4:

- **Real-activity and price series** (GDP, consumption, investment, government, exports, imports, CPI/HICP, all `log`) are benchmarked to IMF World Economic Outlook annual projections (`NGDP_RPCH` real growth, `PCPIPCH` inflation) via temporal disaggregation: a within-year quarterly profile is laid down at the WEO-implied annual growth and then proportional-Denton corrected so that each fully appended year's four-quarter average reproduces the WEO annual level (already-observed quarters in a partial year are held fixed). The realized annual growth of each appended year matches WEO to within 0.2pp.
- **Financial series with no WEO analogue** (NEER, REER, equity indices, and all `lin` rates — short rates, 3-month rates, 10-year yields, unemployment) are carried forward by *hold-last* and flagged as such.

The original raw vintages are left untouched; the extended panels are written separately and consumed by the estimation drivers.

9.6 Predictive Uncertainty Bands

For every bloc the exported baseline carries *predictive* bands, not parameter-only bands. The COVID-missing Gibbs sampler emits a forecast density that integrates over three sources of uncertainty simultaneously — posterior parameter uncertainty, future innovation uncertainty, and missing-data (imputation) uncertainty. The reported quantiles are the 5th, 16th, 84th, and 95th percentiles of this predictive density (with the 50th retained in level space),

$$\big(q_{05},\, q_{16},\, q_{84},\, q_{95}\big)_{T+h} = \text{Quantile}_{\{0.05,0.16,0.84,0.95\}}\big\{\, \tilde{y}^{(d)}_{T+h} \,\big\}_{d=1}^{D}, \tag{57}$$

yielding the 68% band (q_{16}, q_{84}) shaded in the charts and the 90% band (q_{05}, q_{95}). Crucially, the central path $\bar{y}^{\text{baseline}}$ remains the *deterministic* forecast from the posterior-mean coefficients (so the unconditional smoother of Section 4.5 reproduces it exactly, and zero user deviations return the stored baseline); only the bands are predictive. This is the designed fix for the previously over-tight bands that resulted from spreading posterior-mean paths under a tight prior — wide bands now reflect genuine macro volatility (e.g. CN, JP, UK), not the COVID outlier. On-the-fly conditional bands computed in the browser use the same predictive construction, drawing innovations through the simulation smoother of Section 4.6.

9.7 US Specification

The US anchor remains the largest bloc, with $n = 36$ quarterly variables and $p = 3$ lags. Its full variable list is given in Table 2; the fiscal block (`FREV_GDP`, `FPEXP_GDP`, `DDEBT_GDP`) supports the fiscal-reaction counterfactuals in the spirit of Bohn (1998), and the conditional-forecast and optimal-policy machinery of Bańbura et al. (2015) and Barnichon and Mesters (2023) applies uniformly across all blocs.

Table 2: Variable coverage for the US BVAR model. "Trans." denotes the data transformation: `log` = log levels (growth rates displayed), `lin` = levels. "Prior" denotes the prior mean: RW = random walk, WN = white noise.

ID	Description	Trans.	Prior
GDPH	Real Gross Domestic Product	log	RW
CH	Real Personal Consumption Expenditures	log	RW
FRH	Real Private Residential Fixed Investment (nom/deflator)	log	RW
FNH	Real Private Nonresidential Fixed Investment (nom/deflator)	log	RW
XH	Real Exports of Goods & Services	log	RW
MH	Real Imports of Goods & Services	log	RW
GH	Real Government Consumption Expenditures & Gross Investment	log	RW
JGDP	Gross Domestic Product: Chain Price Index	log	RW
JC	Personal Consumption Expenditures: Chain Price Index	log	RW
JCXFE	PCE less Food & Energy: Chain Price Index	log	RW
UI	CPI-U: All Items	log	RW
UIXFDG	CPI-U: All Items Less Food & Energy	log	RW
LXBC	Business Sector: Compensation Per Hour (nominal)	log	RW
LANAGRA	All Employees: Total Nonfarm	log	RW
LR	Civilian Unemployment Rate: 16 yr +	lin	RW
IP	Industrial Production Index	log	RW

ID	Description	Trans.	Prior
CUMFG	Capacity Utilization: Manufacturing [SIC]	lin	RW
HST	Housing Starts	log	RW
YPDH	Real Disposable Personal Income	log	RW
CSENT	University of Michigan: Consumer Sentiment	lin	RW
POLRATE	Effective Federal Funds Rate (policy rate)	lin	RW
FCM2	2-Year Treasury Note Yield at Constant Maturity	lin	RW
FCM5	5-Year Treasury Note Yield at Constant Maturity	lin	RW
FCM10	10-Year Treasury Note Yield at Constant Maturity	lin	RW
FAAA	Moodys Seasoned Aaa Corporate Bond Yield	lin	RW
FBAA	Moodys Seasoned Baa Corporate Bond Yield	lin	RW
FXTWM	Nominal Trade-Weighted Exch Value of US$ vs AFE Currencies	log	RW
SP500	Equity Price Index (OECD US share prices / S&P, public)	log	RW
PZTEXP	Spot Oil Price: West Texas Intermediate	log	RW
GSNE	Non-Fuel Commodity Price Index (IMF, non-energy proxy)	log	RW
SP500VOL	CBOE Volatility Index for S&P 500 Stock Price Index	lin	WN
HP	House Price	log	RW
LEV	Real nonfinancial business debt	log	RW
FREV_GDP	Federal Receipts (% of GDP)	lin	RW
FPEXP_GDP	Federal Primary Expenditure (% of GDP)	lin	RW
DDEBT_GDP	Change in Debt-to-GDP (pp)	lin	RW

The RoW blocs follow the same estimation methodology with country-appropriate variable selections, replacing the US fiscal and detailed financial block with the three `ROW*` stars of Section 9.1; bloc sizes range from roughly $n = 16$ (e.g. CH, NO) to $n = 20$ (EA). The China bloc uses a tighter

Minnesota tightness to discipline its shorter, noisier sample.

10 User Interface Guide

10.1 Top Bar

The top bar provides global controls:

- **Country selector:** Switch between US, EA, UK, JP, CA. Switching reloads all BVAR data and resets conditioning.
- **User Manual:** Download this document as a PDF.
- **Uncertainty mode:** "None" (deterministic), "Quick" ($D = 50$ draws), or "Full" ($D = 200$ draws).
- **Compute button:** Triggers the simulation smoother (Section 4.6).

10.2 Chart Panel

The main panel displays a grid of time-series charts. The first row shows the primary macroeconomic triad—inflation, policy rate, GDP growth—plus government spending. Subsequent rows show core inflation, unemployment, consumption, investment, and any country-specific variables.

Each chart includes:

- A **dotted baseline** forecast with 68% credible bands (shaded).
- A **solid scenario line** showing the conditional forecast.
- A **dashed counterfactual line** (when active) showing the policy counterfactual.
- A **"Condition" button** to toggle conditioning on the variable.
- **Drag interaction:** Click and drag data points on conditioned charts to set target paths. Right-click to toggle individual quarters on or off.

10.3 Control Panel (Sidebar)

10.3.1 Scenario Section

- **Scenario name:** Editable text field.
- **Reset:** Clear all conditioning to return to baseline.
- **Save/Load:** Export and import scenarios as Excel files. The Excel format includes sheets for settings, conditioning paths, active quarters, rate adjustments, and a variable map for reproducibility.

- **Example:** Download a pre-built inflation-surge scenario.
- **Conditioning sliders:** Per-variable, per-quarter deviation sliders (± 3 pp) for each conditioned variable.

10.3.2 Policy Counterfactual Section

- **Counterfactual toggle:** Enter/exit counterfactual mode.
- **DSGE model browser:** Select from 169 models, searchable by name or reference.
- **Policy rule selector:** Choose from 9 standardized monetary policy rules (Table in Section 5.1).
- **Rule-based / Optimal toggle:** Switch between specifying the rate path via a Taylor-type rule or via the Barnichon and Mesters (2023) optimal policy (Section 5.4).
- **Dual-mandate slider:** In optimal mode, adjust the relative weight on price stability versus full employment.

10.4 Export

The scenario bar at the bottom provides four export formats:

- **CSV:** Tabular data with columns for each variable's Q16 (16th percentile), baseline (median), Q84 (84th percentile), active scenario, deviation from baseline, and counterfactual (if active). Rows span historical quarters through the full forecast horizon.
- **Excel:** A workbook with per-variable detail sheets (Q16, baseline, Q84, scenario, deviation, counterfactual) and a summary sheet showing all variables side by side.
- **PNG:** A ZIP archive of high-resolution chart images (1200×500 px) capturing the full visual state including all rendered scenarios, uncertainty bands, and counterfactuals.
- **PDF:** A multi-page PDF with one chart per page (landscape orientation), including metadata footers with country, timestamp, and chart title.

All tabular exports (CSV, Excel) respect the active growth-rate toggle: if year-over-year is selected, the exported values reflect the y/y transformation. Visual exports (PNG, PDF) capture exactly what is displayed on screen, including all visible scenarios and their uncertainty bands.

10.4.1 Multi-Scenario Support

The tool supports up to four simultaneous scenarios. Each scenario's conditional forecast and uncertainty bands are cached when computed, so that switching between scenarios preserves their results. Tabular exports include data for the active scenario; visual exports capture all visible scenario lines and bands.

11 Technical Notes

11.1 Client-Side Computation

All matrix algebra, Kalman filtering, and simulation smoothing are performed in the browser using TypeScript with `Float64Array` storage. The linear algebra kernel implements:

- LU decomposition with partial pivoting for F_t^{-1} (singularity threshold: 10^{-15}).
- Cholesky decomposition for shock draws ($\Sigma = LL'$; diagonal check threshold: 10^{-14}).
- Row-major storage for cache-friendly matrix operations.

11.2 Performance Optimizations

- The unconditional smoother result is cached by a fingerprint of the model dimensions and a sample of coefficient values, avoiding recomputation when only the conditioning paths change.
- Conditioning input is debounced at 30 ms to avoid recomputation during interactive drag operations.
- The 2,500 posterior draws ($\approx$ 44 MB for the US model) are loaded lazily—only when the user clicks "Compute."
- Plotly.js is loaded via dynamic `import()` to avoid blocking initial page render.

11.3 Numerical Conventions

- Measurement noise: $H = 10^{-12}\, I_n$, ensuring the Kalman filter treats conditioned values as near-exact observations while maintaining numerical stability.
- Growth-to-level conversion: $\Delta\ell = \Delta g/4$ for log-transformed variables, reflecting the annualization factor in (2).
- Tikhonov regularization for Barnichon–Mesters: $\alpha = 10^{-6}$.
- All matrix operations use IEEE 754 double precision (64-bit floating point).

12 Discussion

The BVAR Scenario Tool is designed to bridge the gap between the methodological sophistication of conditional forecasting techniques developed in the academic literature and the practical needs of policy economists who require rapid, interactive scenario analysis. We conclude with a discussion of design choices, limitations, and directions for future work.

12.1 Design Philosophy

Several design choices merit discussion. First, the three-layer architecture—baseline, scenario, counterfactual—deliberately separates concerns. The baseline forecast reflects the BVAR's unconditional projection and is computed offline. The scenario layer allows the user to impose conditioning constraints without specifying the structural source of the deviations, following the tradition of Waggoner and Zha (1999). The counterfactual layer then overlays structural identification via DSGE impulse responses, enabling "what-if" policy analysis while preserving the reduced-form covariance structure of the BVAR.

Second, the decision to re-condition the counterfactual through the BVAR (rather than relying solely on the DSGE IRFs) reflects a deliberate methodological choice: the BVAR's estimated covariance structure captures empirical co-movements that may be absent from or imperfectly captured by any single DSGE model. The IRF-consistent mode (Section 5.3c) strikes a balance by using the DSGE model to identify the responses of key variables (inflation, output) to the policy shock, while allowing the BVAR to propagate these effects to the remaining 30+ variables in a data-consistent manner.

Third, all computation runs client-side. This eliminates server infrastructure costs and latency, enables offline use, and avoids transmitting potentially sensitive forecast data over the network. The computational cost is manageable: the Kalman filter and smoother for a 33-variable, 3-lag VAR over a 20-quarter horizon completes in under 50 ms on modern hardware.

12.2 Limitations

The current implementation has several limitations that users should bear in mind:

1. **Linear, time-invariant dynamics:** The VAR coefficients are fixed at their posterior mean for the deterministic conditional forecast. Time-varying parameters, stochastic volatility, and regime switching are not supported. In periods of structural change, the fixed-coefficient assumption may be particularly restrictive.

2. **Gaussian innovations:** The Kalman filter assumes Gaussian innovations. Fat-tailed or asymmetric shock distributions, which may be important during financial crises, are not accommodated.

3. **Point-identified counterfactuals:** The DSGE IRFs used for counterfactual analysis are computed under a single monetary policy rule and model specification. The tool does not average over model uncertainty or rule uncertainty, though the user can manually explore different model–rule combinations.

4. **Lucas critique:** The counterfactual analysis conditions on estimated reduced-form dynamics that may themselves change under the alternative policy. The IRF-consistent mode partially addresses this by using structurally identified responses for the key transmission variables, but

the remaining variables' responses are governed by the estimated BVAR covariance structure, which is held fixed.

5. **Data vintage:** The BVAR is estimated on a fixed data vintage. Real-time data revisions and the publication lag of macroeconomic series are not handled.

12.3 Directions for Future Work

Several extensions are planned or under consideration:

- **Time-varying parameters:** Incorporating stochastic volatility or time-varying coefficients (e.g., Sims and Zha, 1998) to capture evolving dynamics.
- **Structural identification:** Extending the counterfactual layer to support sign restrictions, narrative restrictions, or scenario priors (Farkas and Jakab, 2026) for identification within the BVAR itself, rather than relying on external DSGE models.
- **Mixed-frequency data:** Supporting monthly–quarterly mixed-frequency estimation to incorporate higher-frequency indicators.
- **Model averaging:** Bayesian model averaging over DSGE specifications and policy rules for the counterfactual layer.

Disclaimer

The views expressed herein are those of the author and should not be attributed to the International Monetary Fund, its Executive Board, or its management. This tool is provided "as is" without warranty of any kind, express or implied. The author and the IMF disclaim all liability for any errors, omissions, or for any loss or damage arising from its use. Users are solely responsible for any decisions made based on the output of this tool.

References

Marta Bańbura, Domenico Giannone, and Michele Lenza. Conditional forecasts and scenario analysis with vector autoregressions for large cross-sections. *International Journal of Forecasting*, 31(3): 739–756, 2015.

Regis Barnichon and Geert Mesters. A sufficient statistics approach for macro policy evaluation. *American Economic Review*, 113(11):2809–2845, 2023.

Henning Bohn. The behavior of U.S. public debt and deficits. *The Quarterly Journal of Economics*, 113(3):949–963, 1998.

Lawrence J. Christiano, Martin Eichenbaum, and Charles L. Evans. Nominal rigidities and the dynamic effects of a shock to monetary policy. *Journal of Political Economy*, 113(1):1–45, 2005.

Günter Coenen, Christopher J. Erceg, Charles Freedman, Davide Furceri, Michael Kumhof, René Lalonde, Douglas Laxton, Jesper Lindé, Annabelle Mourougane, Dirk Muir, Susanna Mursula, Carlos de Resende, John Roberts, Werner Roeger, Stephen Snudden, Mathias Trabandt, and Jan in't Veld. Effects of fiscal stimulus in structural models. *American Economic Journal: Macroeconomics*, 4(1):22–68, 2012.

Stephane Dees, Filippo di Mauro, M. Hashem Pesaran, and L. Vanessa Smith. Exploring the international linkages of the euro area: a global VAR analysis. *Journal of Applied Econometrics*, 22(1):1–38, 2007.

Thomas Doan, Robert Litterman, and Christopher Sims. Forecasting and conditional projection using realistic prior distributions. *Econometric Reviews*, 3:1–100, 1984.

James Durbin and Siem Jan Koopman. A simple and efficient simulation smoother for state space time series analysis. *Biometrika*, 89(3):603–616, 2002.

James Durbin and Siem Jan Koopman. *Time Series Analysis by State Space Methods*. Oxford University Press, Oxford, 2nd edition, 2012.

Mátyás Farkas and Zoltán Jakab. Scenario priors for Bayesian SVARs. *Working Paper*, 2026.

Domenico Giannone, Michele Lenza, and Giorgio E. Primiceri. Prior selection for vector autoregressions. *The Review of Economics and Statistics*, 97(2):436–451, 2015.

Domenico Giannone, Michele Lenza, and Giorgio E. Primiceri. Priors for the long run. *Journal of the American Statistical Association*, 114(526):565–580, 2019.

S. Kullback and R. A. Leibler. On information and sufficiency. *The Annals of Mathematical Statistics*, 22(1):79–86, 1951.

Robert B. Litterman. Forecasting with Bayesian vector autoregressions — five years of experience. *Journal of Business & Economic Statistics*, 4(1):25–38, 1986.

M. Hashem Pesaran, Til Schuermann, and Scott M. Weiner. Modeling regional interdependencies using a global error-correcting macroeconometric model. *Journal of Business & Economic Statistics*, 22(2):129–162, 2004.

John C. Robertson, Ellis W. Tallman, and Charles H. Whiteman. Forecasting using relative entropy. *Journal of Money, Credit and Banking*, 37(3):383–401, 2005.

Christopher A. Sims and Tao Zha. Bayesian methods for dynamic multivariate models. *International Economic Review*, 39(4):949–968, 1998.

Frank Smets and Rafael Wouters. Shocks and frictions in US business cycles: A Bayesian DSGE approach. *American Economic Review*, 97(3):586–606, 2007.

John B. Taylor. Discretion versus policy rules in practice. *Carnegie-Rochester Conference Series on Public Policy*, 39:195–214, 1993.

Daniel F. Waggoner and Tao Zha. Conditional forecasts in dynamic multivariate models. *Review of Economics and Statistics*, 81(4):639–651, 1999.

Volker Wieland, Tobias Cwik, Gernot J. Müller, Sebastian Schmidt, and Maik Wolters. A new comparative approach to macroeconomic modeling and policy analysis. *Journal of Economic Behavior & Organization*, 83(3):523–541, 2012.

Volker Wieland, Elena Afanasyeva, Meguy Kuete, and Jinhyuk Yoo. New methods for macro-financial model comparison and policy analysis. In John B. Taylor and Harald Uhlig, editors, *Handbook of Macroeconomics*, volume 2. North Holland/Elsevier, 2016.